

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2062

Prince William Dp 138 kV
August 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2062. The Interconnection Customer proposes a 204.9 MW solar photovoltaic facility interconnecting at the Prince William Dp 138 kV substation in the Virginia area. The request was received on April 12, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 620 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,383,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Fairfax to Jeune Landing 138 KV
Contingency: Toa Vaca to Thomaston 138 KV Line
Loading: 150.7% (Rating: 298 MVA)
DFAX: 5.1%
Pre-project loading: 147.2%
2. Facility: Burke-Sideburn to Yavapai 138 KV
Contingency: Yavapai to Sideburn 138 KV
Loading: 122.9% (Rating: 272 MVA)
DFAX: 6.0%
Pre-project loading: 118.4%
3. Facility: Rocky Reach to Burke-Sideburn 138 KV line
Contingency: Sideburn to Yavapai 138 KV Line
Loading: 119.8% (Rating: 315 MVA)
DFAX: 6.0%
Pre-project loading: 116.0%
4. Facility: Idaho Springs to Rocky Reach 138 KV
Contingency: Sideburn to Yavapai 138 KV
Loading: 118.2% (Rating: 344 MVA)
DFAX: 6.0%
Pre-project loading: 114.6%
5. Facility: W Staunton to Prince William Dp Substation 138 KV
Contingency: Elk Horn to Clover Hill 138 KV L
Loading: 115.8% (Rating: 346 MVA)
DFAX: 55.2%
Pre-project loading: 83.1%

6. Facility: Thomaston Substation to Toa Vaca 138 KV
Contingency: Chaffee Gas Recovery to Blaine 138 KV
Loading: 108.1% (Rating: 412 MVA)
DFAX: 27.6%
Pre-project loading: 94.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fairfax to Jeune Landing 138 KV	Rebuild line with higher-capacity conductor	\$975,000	25% (\$240,000)
Burke-Sideburn to Yavapai 138 KV	Rebuild line with higher-capacity conductor	\$3,875,000	18% (\$688,000)
Rocky Reach to Burke-Sideburn 138 KV line	Rebuild line with higher-capacity conductor	\$1,990,000	18% (\$353,000)
Idaho Springs to Rocky Reach 138 KV	Rebuild line with higher-capacity conductor	\$1,180,000	18% (\$209,000)
W Staunton to Prince William Dp Substation 138 KV	Rebuild line with higher-capacity conductor	\$839,000	78% (\$657,000)
Thomaston Substation to Toa Vaca 138 KV	Reconductor line	\$821,000	29% (\$236,000)
	Total allocated to project		\$2,383,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.